2nd WMO-IMO Symposium on Extreme Maritime Weather

Bridging the Knowledge Gap Towards Safer Shipping

● 23-26 September 2024 ● IMO Headquarters, London

Provisional Programme



NAVIGATING THE FUTURE: SAFETY FIRST!

IMO INTERNATIONAL MARITIME ORGANIZATION

WORLD METEOROLOGICAL ORGANIZATION

Early Warnings for All

2021 2030 United Nations Decade of Ocean Science for Sustainable Development

Monday, 23 September 2024		
Time	**Session**	**Objectives / topics**
9:30 – 10:30	Session 1: **Opening Session and Welcome Remarks**	• Welcome / Practical Information. • Opening Remarks. • Updates since the 1st Symposium.
10:30 – 11:00	Coffee Break	
11:00 – 12:30	Session 2: **Extreme Weather and Safety at Sea**	**Objective:** Discuss recent cases of extreme and hazardous weather events and the impacts of climate change on maritime activities, highlighting the need for enhancing collaboration among the metocean, maritime, insurance, and other relevant communities and stakeholders for safety at sea. • Overview of recent extreme and hazardous maritime weather events and the concept of meteorological maritime safety information. • Recent indications in weather related maritime incidents: Are better forecasts reducing loss of life and property at sea? • Perspectives from the insurance industry. • Impacts of extreme weather and climate change on port infrastructure.
12:30 – 14:00	Lunch	
14:00 – 15:00	Session 3: **Enhancing Knowledge Opportunities for Seafarers and Forecasters**	**Objective:** Highlight the state of training in marine weather for both seafarers and metocean forecasters and consider possible enhancements to training programmes. • Perspectives from the: ○ meteorological community ○ maritime community ○ maritime training community.
15:00 – 15:30	Tea Break	
15:30 – 17:00	**Interactive Scenario**	
17:00 – 17:30	**Day 1 Summary**	
18:00 – 20:00	**WMO-IMO Evening Reception**	

Tuesday, 24 September 2024		
Time	**Session**	**Objectives / topics**
9:30 – 11:00	Session 4: **Observations and Data Collection**	**Objective:** Provide an overview of current weather and ocean observations at sea, highlighting the complementary nature of ship observations with other ocean observing systems and their impact on the quality of maritime weather early warnings and forecasts. • Value of the ocean observing system. • Forecaster perspective on ship observations. • Overview of the Voluntary Observing Ship (VOS) Scheme. • Seafarers view of ship observations. • Future of shipboard observations.
11:00 – 11:30	Coffee Break	
11:30 – 12:30	**Panel Discussion**	
12:30 – 14:00	Lunch	***Side Event: Metocean forecasting support in polar regions (45 minutes)***
14:00 – 15:30	Session 5: **Dissemination of MetOcean Information to Ships**	**Objective:** Discuss the status of and possibilities for improving dissemination of metocean early warnings and forecasts to vessels and the maritime community, enabling seafarers to make sound routeing and other operational decisions at sea. • Status of S-41X sea ice and weather overlays in S-100 based product specifications for Electronic Chart Display and Information Systems. • SMART-Navigation: Implementing e-Navigation for Ships. • Dissemination of MSI in the Polar Regions - Satellite coverage and user experience. • Use of emerging technologies to support dissemination of MSI at sea.
15:30 – 16:00	Tea Break	
16:00 – 17:00	**Panel Discussion**	
17:00 – 17:30	**Day 2 Summary**	

Wednesday, 25 September 2024		
Time	**Session**	**Objectives / topics**
9:30 – 11:00	Session 6: **Better Products and Services for Improved Weather Warnings and Forecasts**	**Objective:** Explore how metocean products and services, including early warnings and services, can better meet seafarers' current and future needs to support safe and efficient maritime operations. ● Optimizing ship routeing for safety, fuel efficiency, and reduced emissions. ● Early warnings and impact-based forecasting for maritime safety. ● Metocean support for emergencies at sea, including both safety and environment related, and search and rescue. ● Products and services for cruise ships and tourism: how public and private sector providers support forecasts at sea.
11:00 – 11:30	Coffee Break	
11:30 – 12:30	**Panel Discussion**	
12:30 – 14:00	Lunch	
14:00 – 15:30	Session 6 (Cont.)	● Metocean Information for SOLAS and Polar Code Requirements: Are user needs being met? ● Enhancing metocean support for non-SOLAS ships. ● Weather safety for domestic ferry operations. ● Role of non-government service providers in support of maritime operations.
15:30 – 16:00	Tea Break	
16:00 – 17:00	**Panel Discussion**	
17:00 – 17:30	**Day 3 Summary**	

Thursday, 26 September 2024		
Time	**Session**	**Objectives / topics**
9:30 – 11:00	Session 7: **Strengthening Collaboration**	**Objective:** Discuss how to improve the dialogue across all sectors of the maritime and metocean communities to enhance safety at sea and strengthen continued collaboration after the Symposium. **Speaking topics:** Speakers from various relevant maritime related organizations will discuss how improved collaboration in maritime weather support can better support their members. Similarly, representatives of meteorological agencies from various regions will discuss how they work within their nation/region to connect to authorities and work with users, as highlighted through examples specific to their geographic area.
11:00 – 11:30	Coffee break	
11:30 – 12:30	Session 8: **Bridging the Knowledge Gap Towards Safer Shipping - Summary Conclusion**	**Objective:** Summary of key outcomes and issues emerging from the Symposium and looking to the future.
12:30	**Closing of Symposium**	